



EUROPEAN COMMISSION 2026

Covariates in DiD

OLS specs, IPW, OR, DR, and why the spec matters

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Covariates fix broken parallel trends — but the spec has to be right

The standard TWFE with additive X :

$$Y_{it} = \alpha_i + \lambda_t + \delta (D_i \times \text{Post}_t) + \beta X_i + \varepsilon_{it}$$

This imposes **two strong hidden assumptions**:

1. The slope of X on $Y(0)$ is the **same in pre and post**. (No time-varying returns.)
2. The treatment effect δ is **the same for every value of X** . (Homogeneous TE in X .)

If either assumption fails, adding X additively gives you the wrong answer. Specification is not a cosmetic choice.

Worry 1: the effect of X on $Y(0)$ changes over time

Problem. β in the additive model is a single number. But returns to education, age, race gap etc. can shift at the same moment as your policy. That shift looks exactly like a treatment effect.

Fix: interact Post with X .

$$Y_{it} = \alpha_i + \lambda_t + \delta (D_i \times \text{Post}_t) + \beta X_i + \gamma (\text{Post}_t \times X_i) + \varepsilon_{it}$$

γ picks up the *change* in the slope of X between pre and post.

The ATT estimate $\hat{\delta}$ is now protected against returns-to- X shifting at the policy break.

This is equivalent to interacting every covariate with a time dummy. Cost: $(k + 1)$ extra parameters for k covariates.

Worry 2: the treatment effect itself varies with X

Problem. If $\delta_i = \delta(X_i)$, the additive spec estimates a *variance-weighted* average of $\delta(X)$ — not the ATT. The weights depend on X 's variation across the sample, not on the share treated.

Fix: interact D (the treatment indicator) with X .

$$Y_{it} = \alpha_i + \lambda_t + \delta_0 (D_i \times \text{Post}_t) + \tau (D_i \times \text{Post}_t \times X_i) + \beta X_i + \varepsilon_{it}$$

Now $\delta_0 + \tau X_i$ is the treatment effect for a unit with covariate value X_i .

The ATT is recovered by averaging $\hat{\delta}_0 + \hat{\tau} X_i$ over the *treated* sample.

Standard in heterogeneous-TE DiD (e.g. Callaway-Sant'Anna with covariates). In practice, use `drdid` or `csdid` rather than hand-coding this.

Both worries at once: the fully saturated spec

Include *all three* interaction blocks:

$$Y_{it} = \alpha_i + \lambda_t + \underbrace{\delta_0 (D_i \times \text{Post}_t)}_{\text{avg TE}} + \underbrace{\tau (D_i \times \text{Post}_t \times X_i)}_{\text{TE varies in } X} + \underbrace{\gamma (\text{Post}_t \times X_i)}_{\text{returns vary over time}} + \beta X_i + \varepsilon_{it}$$

Recovery of the ATT:

$$\widehat{\text{ATT}} = \frac{1}{n_T} \sum_{i: D_i=1} (\hat{\delta}_0 + \hat{\tau} X_i)$$

This is Spec C in the lab do-file (“Fully Saturated TWFE”).

Worries 1 and 2 together exhaust the ways an additive OLS spec can fail. The non-parametric fix is to leave OLS entirely and use OR, IPW, or DR.

IPW — Abadie (2005): reweight the controls

Intuition. Weight the control group so it *looks like the treated group* in X . Then the reweighted control trend is a valid counterfactual.

Estimator.

$$\widehat{\text{ATT}}_{\text{IPW}} = \frac{1}{n} \sum_{i=1}^n \underbrace{\frac{D_i - \hat{p}(X_i)}{1 - \hat{p}(X_i)}}_{\text{IPW weight}} \cdot \Delta Y_i \bigg/ \bar{D}$$

where $\hat{p}(X_i) = \widehat{\text{Pr}}(D_i = 1 \mid X_i)$ is the **propensity score** and $\Delta Y_i = Y_{i,\text{post}} - Y_{i,\text{pre}}$.

Five steps.

1. **Specify and estimate the propensity score.** Regress D_i on X_i (logit or probit).
2. **Check overlap.** Plot $\hat{p}(X)$ by treatment status. Trim if needed.
3. **Construct weights.** $w_i = (D_i - \hat{p}(X_i)) / (1 - \hat{p}(X_i))$.
4. **Take the weighted average.** $\widehat{\text{ATT}}_{\text{IPW}} = (\sum w_i \Delta Y_i) / n_T$.
5. **Standard error.** Bootstrap or influence-function SE.

Step 2 is non-negotiable: check overlap

IPW weights blow up when $\hat{p}(X_i) \rightarrow 1$ for control units.

- **Common support:** $\hat{p}(X_i) \in [\epsilon, 1 - \epsilon]$ for all i . If some treated units have $\hat{p} \approx 1$, no control unit resembles them — you have no counterfactual.
- **Diagnosis:** histogram of $\hat{p}$ separately for $D = 0$ and $D = 1$. They should substantially overlap.
- **Fix:** trim units outside support (Crump et al. 2009: keep $\hat{p} \in [0.1, 0.9]$) or use a trimmed ATT.

If the propensity score separates treated from controls perfectly, IPW is undefined. No method fixes lack of overlap — it is a *data* problem, not an estimation problem.

OR — Heckman, Ichimura & Todd (1997): impute the counterfactual

Intuition. Model the *trend* of untreated potential outcomes conditional on X . Use that model to predict what the treated group *would have done* absent treatment.

Estimator.

$$\widehat{\text{ATT}}_{\text{OR}} = \frac{1}{n_T} \sum_{i: D_i=1} [\Delta Y_i - \hat{\mu}_0(X_i)]$$

where $\hat{\mu}_0(X_i) = \widehat{E}[\Delta Y \mid D = 0, X]$ is the **outcome model** — the predicted trend for a control unit with characteristics X_i .

Steps.

1. **First-difference.** Compute $\Delta Y_i = Y_{i,\text{post}} - Y_{i,\text{pre}}$ for all units.
2. **Fit the outcome model.** Regress ΔY on X using controls only ($D_i = 0$).
3. **Predict counterfactuals.** Evaluate $\hat{\mu}_0(X_i)$ for every treated unit.
4. **Take the mean gap.** $\text{ATT} = \overline{\Delta Y}^T - \overline{\hat{\mu}_0(X)}^T$.
5. **Standard error.** Bootstrap.

Doubly robust — Sant’Anna & Zhao (2020)

Intuition. Combine OR and IPW. One model can be misspecified and the estimator is still consistent — but not both.

Estimator.

$$\widehat{\text{ATT}}_{\text{DR}} = \frac{1}{\bar{D}} \frac{1}{n} \sum_{i=1}^n \underbrace{\frac{D_i - \hat{p}(X_i)}{1 - \hat{p}(X_i)}}_{\text{IPW correction}} \cdot \underbrace{[\Delta Y_i - \hat{\mu}_0(X_i)]}_{\text{OR residual}}$$

- If $\hat{\mu}_0$ is correct but $\hat{p}$ is wrong $\Rightarrow$ consistent. The OR residuals are mean-zero for the controls, so misweighting doesn’t matter.
- If $\hat{p}$ is correct but $\hat{\mu}_0$ is wrong $\Rightarrow$ consistent. IPW balances X , so the misspecified trend washes out.
- If *both* are wrong $\Rightarrow$ biased. “Doubly robust” $\neq$ doubly unbeatable.

Implementation: `drdid re $X, time(year) ivar(id) tr(ever_treated) dripw` in Stata.

Lab: run all estimators on the LaLonde data

Open the do-file:

```
Labs/Lalonde-Covariates/lalonde_all_specs.do
```

The script runs **ten estimators** on the LaLonde–Dehejia-Wahba non-experimental panel (NSW treated + CPS controls) and compares each to the RCT benchmark of **\$1,794**:

- Spec 0: Naive TWFE (no X)
- Spec A: Additive X
- Spec B: Post $\times X$
- Spec C: Saturated TWFE (FD)
- OR by hand (HIT 1997)
- OR via drdid regadjust
- IPW by hand (Abadie 2005)
- IPW via drdid ipw
- DR by hand (Sant'Anna-Zhao)
- DR via drdid dripw

Data: lalonde_nonexp_panel.dta. Pre = 1975, Post = 1978. Covariates: age, education, marital status, race, re74, u74.

A Monte Carlo to settle the rankings

- Compare four estimators on the same data: **TWFE** (with X as additive controls), **OR**, **IPW**, **DR**.
- **4 DGPs** (Sant'Anna's design): vary whether the OR model and the pscore model are correctly specified.
- **Sample size:** 1,000 units. **Replications:** 10,000.
- **Report:** bias, RMSE, SE, 95% coverage, CI length.

The question we want answered: when does DR pay off, and when does it break?

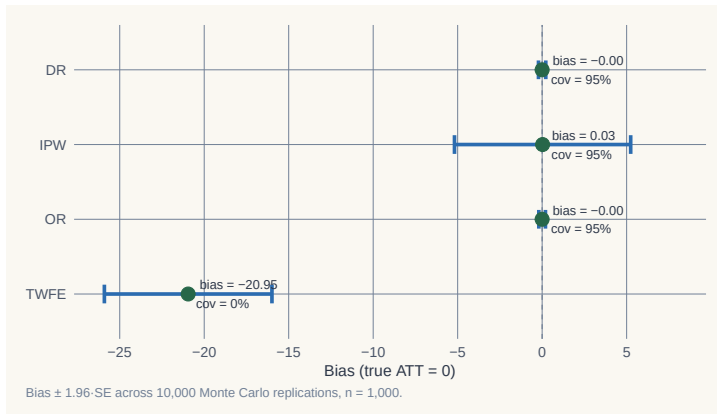
Driver code: `simulations/covariates/covariates_monte_carlo.{do,R}`

DGP 1 — both models correct

	Bias	RMSE	SE	Coverage	CI length
TWFE	−20.95	21.12	2.53	0.000	9.91
OR	−0.001	0.10	0.10	0.950	0.40
IPW	+0.026	2.77	2.66	0.952	10.44
DR	−0.001	0.11	0.11	0.947	0.41

*TWFE catastrophic (Bias ≈ -21 ; coverage zero). OR & DR both efficient (Bias ≈ 0 , coverage 95%). IPW unbiased but **wide CIs**.*

DGP 1 — visualizing the sampling distribution



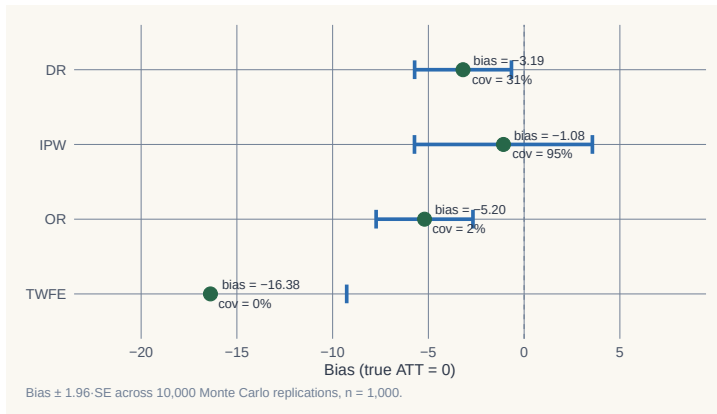
When both models are correct, OR and DR concentrate tightly on the truth. IPW is unbiased but noisy. TWFE is off the map.

DGP 4 — both models *wrong*

	Bias	RMSE	SE	Coverage	CI length
TWFE	−16.38	16.54	3.63	0.000	14.22
OR	−5.20	5.36	1.29	0.015	5.05
IPW	−1.08	2.66	2.37	0.949	9.31
DR	−3.19	3.45	1.29	0.308	5.07

*DR is not magic. When both models fail, DR is biased and undercovers (0.31 vs. nominal 0.95).
IPW alone now wins on coverage — it doesn't rely on the outcome model.*

DGP 4 — the sampling distribution tells the same story



DR no longer hugs the truth. IPW is wider but centered. Doubly robust $\neq$ doubly unbeatable.

What the Monte Carlo teaches

- **TWFE with additive X :** never wins. Coverage is zero in every DGP. **Stop using it as a default.**
- **OR:** efficient when the outcome model is right. Disaster when it's wrong (Bias -5 , coverage 1.5%).
- **IPW:** less efficient but *robust to outcome misspecification*. The unsexy workhorse.
- **DR:** best of both *when at least one is right*. Not better than IPW when *both* are wrong.

“Doubly robust” means robust to one misspecification. Not two. Not magic.

The lesson: specification is not cosmetic

All five estimators today require you to *correctly specify something*:

- **OLS specs (A, B, C):** you must commit to how X enters $Y(0)$ *and* whether δ is homogeneous in X . Wrong choice $\Rightarrow$ wrong ATT.
- **OR:** the outcome model $\hat{\mu}_0(X)$ must be correctly specified.
- **IPW:** the propensity score $\hat{p}(X)$ must be correctly specified, and overlap must hold.
- **DR:** *at least one* of the two must be right.

Adding covariates does not make your DiD credible. It transfers the credibility question from “does PT hold?” to “is my model of $Y(0)$ or $p(X)$ correctly specified?” You still have to answer it.

The question is always
“did I get the model right?”
